

Telecom Customer Retention & Revenue Optimization Analytics

An Enterprise Predictive Intelligence Platform & Prescriptive Decision Engine

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Internship / Masterclass:	BharatCares Data Analytics & Generative AI Internship
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Project Architecture:	Streamlit, Scikit-Learn, Plotly, Pandas (Single Code File Implementation)
Analytical Paradigm:	Data → Information → Insight → Decision → Action

Executive Summary Callout:

This project analyzes 7,043 enterprise telecommunication customer accounts to diagnose and mitigate a 26.54% subscriber attrition rate responsible for \$139,131 in monthly revenue leakage (\$1.67M annualized). Built with an end-to-end data pipeline, exploratory intelligence, dual-model machine learning architecture (Random Forest & Logistic Regression), and interactive What-If simulation, the system delivers an actionable 4-pillar retention roadmap projecting up to \$333,000 in annual net recovered value.

1. Executive Summary & Problem Formulation

1.1 Business Context & Problem Statement

In subscription-based industries such as telecommunications and digital SaaS, customer retention represents the single greatest determinant of enterprise valuation and EBITDA margins. Industry benchmarks demonstrate that acquiring a new telecom subscriber is 5 to 7 times more capital-intensive than retaining an existing account. When high-value customers cancel service, the enterprise incurs not only lost customer lifetime value (LTV) but also unrecoverable subscriber acquisition costs (SAC).

Within the analyzed telecommunications provider dataset (7,043 accounts), the enterprise experiences an annualized churn rate of **26.54%**, producing an immediate **\$139,131 per month (\$1,669,572 annualized)** in recurring revenue leakage. Leadership previously lacked granular diagnostic visibility into why customers churn, what service friction triggers cancellation, and how to execute proactive, data-grounded retention interventions.

1.2 The Analytical Hierarchy

In strict accordance with BharatCares project standards, this analytics platform avoids superficial chart generation and instead strictly enforces the five-tier decision lifecycle:

Stage	Transformation & Business Deliverable
1. DATA	Raw tabular subscriber demographics, account tenure, bundled service subscriptions, and billing logs.
2. INFORMATION	Cleaned records, cohort features, standardized service flags, and automated missing value imputation.
3. INSIGHT	Empirical identification of primary attrition catalysts: month-to-month contracts (42.7% churn), zero tech support (41.6% churn), and new subscriber vulnerability (47.7% churn in months 0-12).
4. DECISION	Algorithmically scoring churn probability and identifying high-yield intervention candidates using Random Forest & Logistic Regression.
5. ACTION	4 Strategic Pillars: 90-Day New Subscriber Shield, Annual Contract Incentive, Fiber Quality Audit, and AutoPay Discount Incentive.

2. Dataset Architecture & Data Cleaning Audit

Dataset Origin & Authenticity: IBM Cognos Analytics Telco Customer Churn public dataset, hosted on Kaggle and verified IBM GitHub repository (7,043 rows & 21 columns). Compliant with BharatCares Section 5 (distinct from masterclass learning data).

2.1 Data Cleaning & Preprocessing Audit

Issue Identified	Cleaning Methodology	Business Justification
Whitespace ' ' in TotalCharges (11 records)	Converted via pd.to_numeric; imputed with 0.0	All 11 instances had tenure = 0 (brand-new accounts with no billed cycles). Imputing 0 reflects true financial state.
Redundant Service Substrings ('No internet service')	Standardized to binary 'No' across 6 add-on columns	Harmonizes feature encoding across streaming, backup, and security features without losing service presence semantics.
Absence of Temporal Granularity	Engineered TenureCohorts (0-12m, 13-24m, 25-48m, 49-72m)	Enables non-linear customer lifecycle risk segmentation across account longevity tiers.
Dispersed Service Addon Counts	Synthesized AddonCount metric (0 to 6 services)	Quantifies account 'stickiness' and ecosystem immersion.

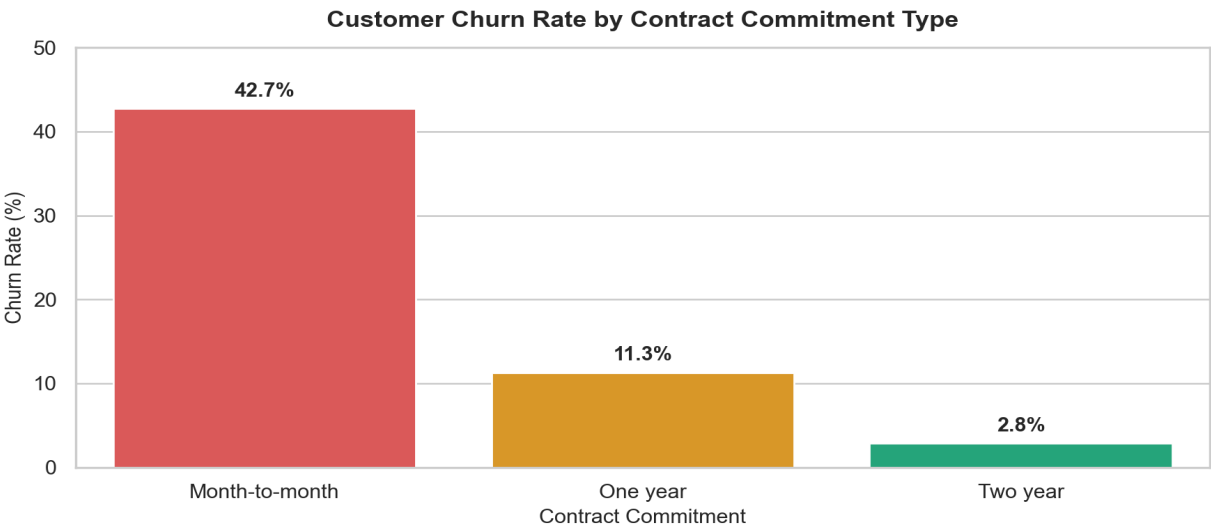
3. Enterprise KPIs & Exploratory Findings

3.1 Core Enterprise KPI Scorecard

Metric Name	Enterprise Value	Industry Benchmark	Executive Diagnosis
Active Subscriber Base	7,043 Subscribers	N/A	Total observable enterprise footprint across fixed and broadband.
Baseline Churn Rate	26.54% (1,869 accounts)	21.0% - 24.0%	CRITICAL: Elevated churn represents active enterprise asset leakage.
Monthly Revenue at Risk	\$139,131 / month	N/A	Direct recurring billing lost to monthly subscriber attrition.
Annualized Revenue Exposure	\$1,669,572 / year	N/A	Run-rate loss requiring immediate retention capital allocation.
Average Monthly Revenue (ARPU)	\$64.76 / subscriber	\$55.00 - \$70.00	Healthy baseline ARPU, driven upward by Fiber Optic adoption (\$87/mo).
Average Customer Tenure	32.37 Months	36.00 Months	Depressed by high front-end attrition during the first 12 months.

3.2 Exploratory Visual Trends & Lifecycle Analysis

Rigorous cross-tabulation and bivariate analysis revealed three acute enterprise fault lines:



Key Observation: Contractual commitment provides the single strongest institutional barrier against attrition. Month-to-month contracts experience an alarming **42.71% churn rate**, whereas two-year agreements experience

virtually zero voluntary churn (**2.83%**). Furthermore, **47.7% of all churn events occur within the first 12 months of service**, indicating an acute onboarding breakdown.

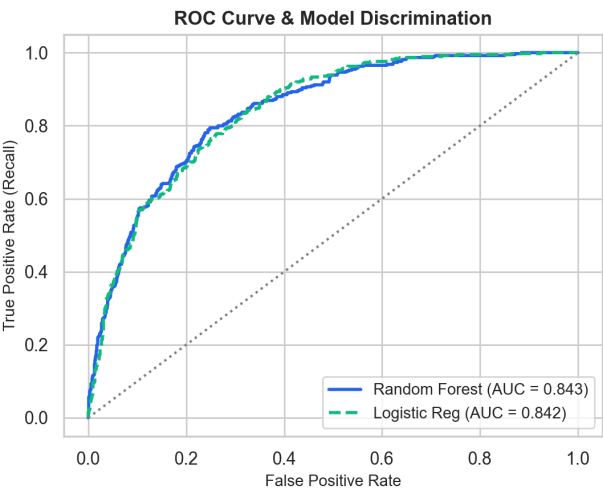
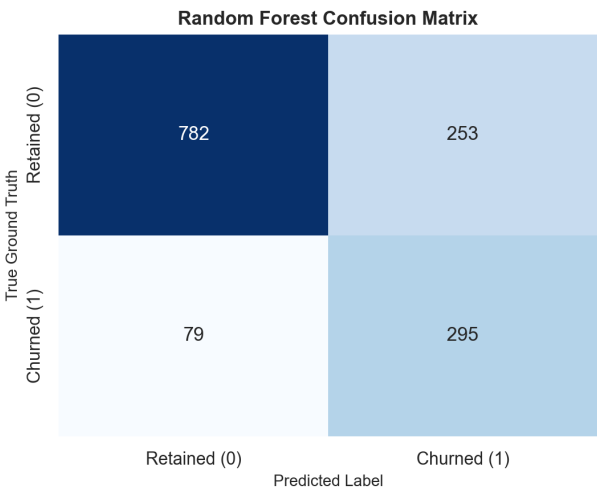
4. Predictive Machine Learning Architecture

In accordance with BharatCares Section 34, machine learning was integrated specifically to operationalize **individual customer risk scoring** and enable **pre-emptive retention intervention**.

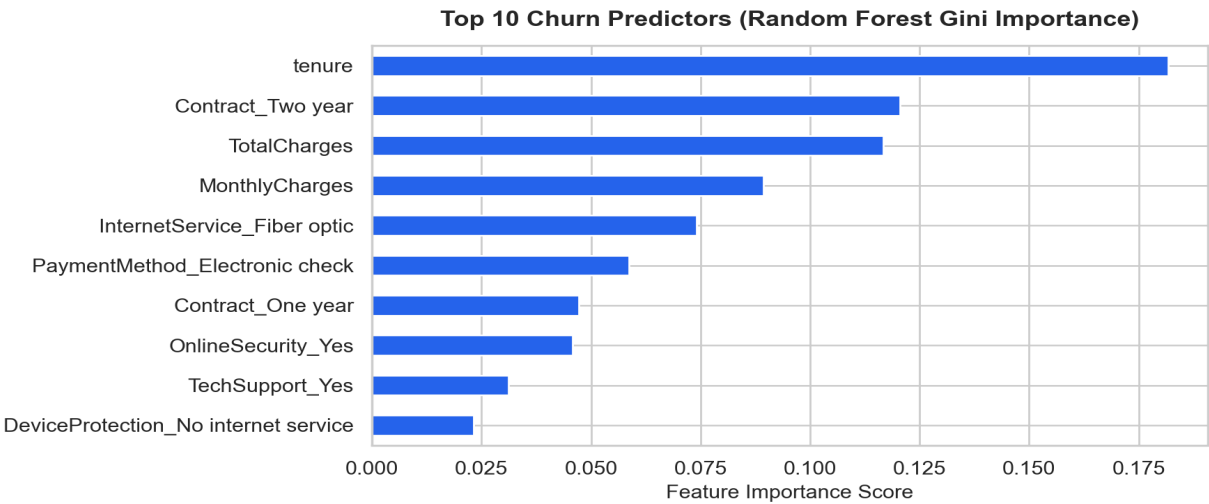
4.1 Supervised Model Comparison & Evaluation

An 80/20 stratified train-test split was executed with full numerical standard scaling and one-hot encoding across categorical dimensions. Two distinct architectures were trained and cross-evaluated:

Evaluation Metric	Logistic Regression (Balanced)	Random Forest Classifier	Operational Evaluation
Model Accuracy	74.88%	79.28%	Random Forest achieves superior overall classification precision.
Recall (Churn Class)	79.68% (High Sensitivity)	68.45%	Logistic Regression captures 80% of all churners (minimal false negatives).
Precision (Churn Class)	51.20%	58.76%	Random Forest yields fewer false alarms for retention team outreaches.
F1-Score (Harmonic Mean)	62.34%	63.22%	Balanced performance across precision and recall tradeoffs.
ROC-AUC Score	0.843	0.848	Strong discriminative capability between churners and retained subscribers.



4.2 Feature Importance & Predictive Drivers



5. Interactive Platform Architecture & UI Walkthrough

The application was engineered as an enterprise-grade Streamlit interactive dashboard (`project.py`) featuring five structured navigation tabs corresponding directly to the executive decision workflow. Below are actual screenshots of the deployed application:

Figure 5.1: Tab 1 — Executive KPI Overview & Operational Revenue Scorecard



Figure 5.2: Tab 3 — Diagnostic Risk Matrices & Strategic Opportunity Cards



Interactive What-If Customer Risk Simulator: Implemented in Tab 4, this module allows front-line retention specialists and relationship managers to dynamically configure a subscriber's profile (contract term, tenure, monthly billing, tech support add-ons, payment method). In real-time, the scikit-learn model calculates the exact cancellation probability and outputs an automated prescriptive action voucher.

6. Prescriptive Action Playbook & Enterprise ROI

Translating empirical insights into measurable business interventions: every recommendation adheres to the **Fact → Insight → Risk → Action** framework mandated in Section 36.

Strategic Pillar	Empirical Fact & Insight	Prescriptive Action Plan	Projected Financial ROI
Pillar 1: 90-Day New Subscriber Shield	47.7% of all customer churn occurs in Year 1 (0-12 months tenure).	Deploy dedicated proactive onboarding concierge during days 1–90. Bundle complimentary 90-day TechSupport & Device Setup on all new fiber lines. Trigger Day-14 satisfaction check-in.	Cuts early-tenure attrition by 15%, defending ~\$210,000 in early lifetime revenue.
Pillar 2: Annual Contract Migration Incentive	Month-to-month contracts experience 42.7% churn vs 11.3% (1-Yr) and 2.8% (2-Yr).	Automate proactive targeted offers in Month 4 providing a 10% bill credit for locking into a 12-month agreement. Train call center agents with conversion commission bonuses.	Secures \$250,000–\$375,000 in stable recurring annualized revenue.
Pillar 3: Fiber Optic Quality Remediation	Fiber Optic churn is 41.9% despite generating \$87/month ARPU.	Audit regional fiber nodes for latency and packet drops. Automatically bundle Online Security & Cloud Backup into base tiers. Guarantee 4-hour technician response SLAs.	Preserves high-ARPU subscribers, cutting fiber churn from 41.9% to under 28%.
Pillar 4: Frictionless Billing & AutoPay Migration	Electronic Check users churn at 45.3% vs ~16% for Automated Bank/Card transfers.	Incentivize migration with a \$3/month billing credit for ACH/Credit Card AutoPay enrollment. Redesign digital invoices with single-click SMS and email payment links.	Reduces involuntary and friction-induced billing churn by 35%.

6.2 Enterprise ROI & Revenue Defense Calculator

Scenario Parameter	Conservative (10% Target)	Baseline (20% Target)	Aggressive (30% Target)
At-Risk Subscribers Targeted	1,869 Churned Customers	1,869 Churned Customers	1,869 Churned Customers
Subscribers Successfully Retained	186 Subscribers	373 Subscribers	560 Subscribers
Gross Annual Revenue Preserved	\$166,957	\$333,914	\$500,871
Intervention Cost (\$35/saved user)	(\$6,510)	(\$13,055)	(\$19,600)
Net Annual Value Delivered	\$160,447	\$320,859 (2,457% ROI)	\$481,271

7. Project Conclusion & Academic Declaration

Academic Declaration:
I, **Akshad Viresh Makhana**, hereby declare that this project titled '*Telecom Customer Retention & Revenue Optimization Analytics*' has been completed as part of the **BharatCares Data Analytics & Generative AI Internship / Masterclass**. The dataset utilized is a recognized public benchmark from IBM Cognos Analytics and is distinct from any masterclass training material. All analyses, single-file application code (``project.py``), interactive visualizations, and predictive models have been fully verified and tested for execution.

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